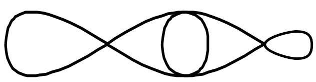
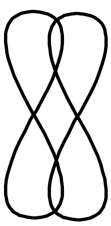
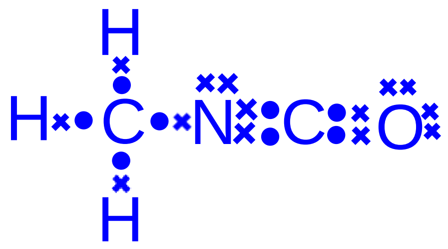
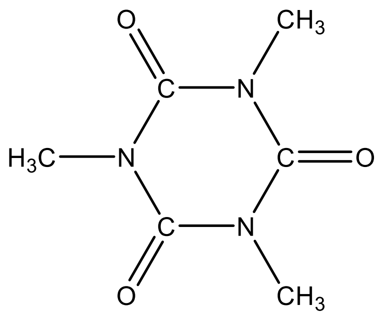
**RIVER VALLEY HIGH SCHOOL**
**H2 CHEMISTRY**

## Suggested Solutions for 2018 A Level H2 Chemistry Paper
## Paper 1

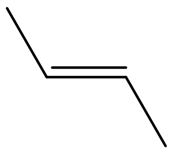
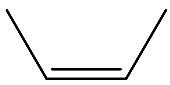
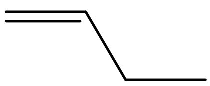
| *Question Number* | *Key* | *Question Number* | *Key* |
|---|---|---|---|
| 1 | **D** | 16 | **A** |
| 2 | **D** | 17 | **B** |
| 3 | **D** | 18 | **A** |
| 4 | **B** | 19 | **C** |
| 5 | **C** | 20 | **D** |
| 6 | **D** | 21 | **C** |
| 7 | **C** | 22 | **A** |
| 8 | **C** | 23 | **B** |
| 9 | **C** | 24 | **D** |
| 10 | **B** | 25 | **A** |
| 11 | **C** | 26 | **B** |
| 12 | **D** | 27 | **B** |
| 13 | **A** | 28 | **C** |
| 14 | **A** | 29 | **D** |
| 15 | **A** | 30 | **D** |

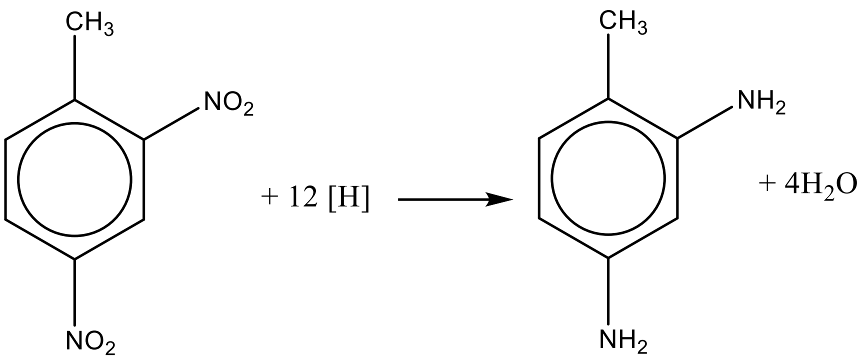
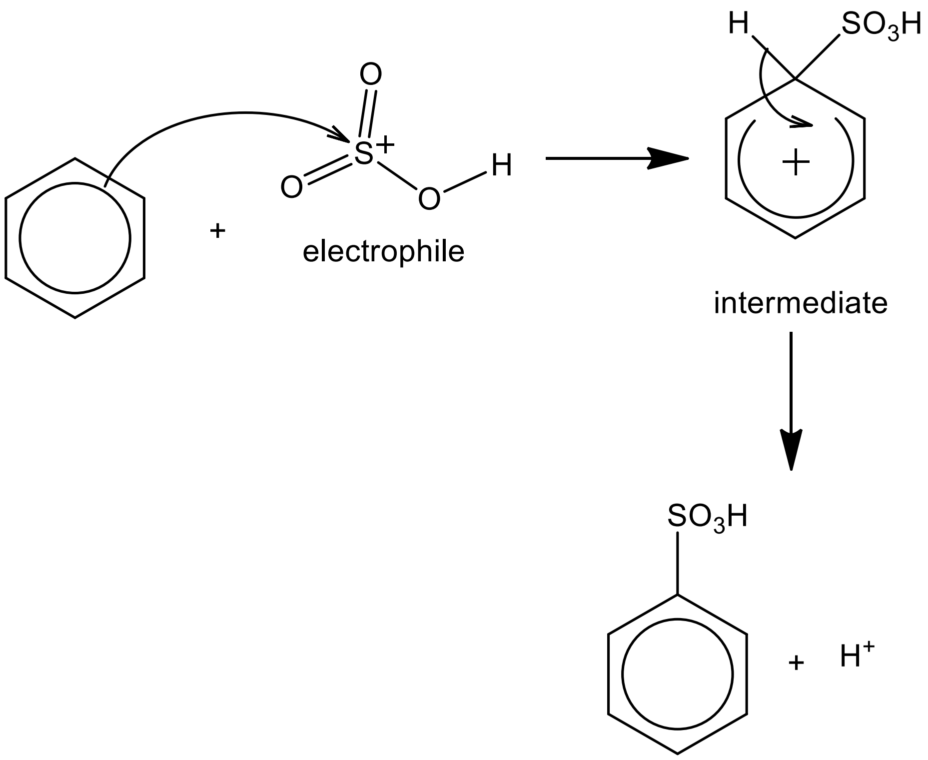
**Paper 2**

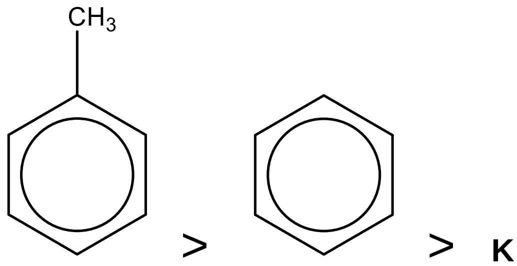
<table>
<tr><td>1</td><td>(a)</td><td>(i)</td><td>
<table>
<tr><th></th><th>similarity</th><th>difference</th></tr>
<tr><td>1s and 2s orbital</td><td>spherical in shape</td><td>1s orbital is smaller (less diffused) and closer to the nucleus than 2s orbital</td></tr>
<tr><td>2s and 2p orbital</td><td>Both orbitals belong to the second principal quantum shell.<br>The probability of finding electrons reaches zero at the nucleus.</td><td>2s is spherical but 2p is dumb-bell in shape</td></tr>
</table>
</td></tr>
<tr><td></td><td></td><td>(ii)</td><td>The C atom in phosgene has 3 bond pairs (pairs of bonding electrons) and 0 lone pair (pair of non-bonding electrons). As a result of VSEPR, to minimise the repulsion between the electron pairs, the shape around the C atom will be trigonal planar, with a bond angle of 120º. The shape of the phosgene molecule is also trigonal planar.</td></tr>
<tr><td></td><td></td><td>(iii)</td><td>$\sigma$ bonds<br><br>2p (O)    $sp^2$ (C)<br>$\pi$ bond<br><br>2p (C)    2p (O)</td></tr>
<tr><td></td><td>(b)</td><td>(i)</td><td>Electronegativity of an atom is a measure of its ability to attract electrons in a covalent bond. The higher the electronegativity, the stronger it will attract and pull the bonding electrons towards itself.</td></tr>
</table>

| | | | |
|---|---|---|---|
| | | **(ii)** | There can be permanent dipole-permanent dipole interactions (pd-pd) and instantaneous dipole-induced dipole interactions (id-id) between the phosgene molecules.<br><br>As the C atom is bonded to the more electronegative C*l* and O atoms, and O is more electronegative than C*l*, the polar molecule will have (dipoles which do not cancel out to give) a net dipole moment, with pd-pd interactions between the molecules.<br><br>In the phosgene molecule which has a simple covalent structure, due to the instantaneous distortion of the electron cloud in the molecule, an instantaneous dipole can be formed which will induce a temporary dipole in another molecule. This will give rise to id-id interaction between the molecules. |
| | **(c)** | **(i)** |  |
| | | **(ii)** |  |
| | | **(iii)** | **A** has a larger electron cloud than methyl isocyanate, thus more energy is needed to overcome the stronger instantaneous dipole-induced dipole (id-id) interactions between molecules of **A** than the weaker permanent dipole-permanent dipole interactions between methyl isocyanate molecules.<br><br>(Strong responses were specific in using intermolecular forces to explain the difference in physical state.) |

| | | | |
|---|---|---|---|
| **2** | **(a)** | Dynamic equilibrium refers to a reversible process at equilibrium in which the rate of forward reaction is equal to the rate of backward reaction. | |
| | **(b)** | $K_p = \dfrac{p_{NO}{}^2}{p_{N_2} p_{O_2}}$ | |
| | **(c)** | Positive. An increase in temperature favours the endothermic reaction to absorb some heat. As temperature increases, magnitude of $K_p$ increases which indicates that the position of equilibrium shifts right to favour the forward reaction. Since the forward reaction is favoured, the forward reaction is endothermic. | |
| | **(d)** | **(i)** | As the volume expands from 1 $dm^3$ to 5 $dm^5$, the partial pressures for all individual gases will decrease to 1/5 of the original. Since the number of reactant and product gases are the same (2 moles on each side of the equation), the decrease in partial pressures of the gases will not cause the position of equilibrium to shift and the system remains at equilibrium. Therefore, the composition of the reaction mixture will not be affected apart from the decrease in concentration. |
| | | **(ii)** | $K_p$ remains the same. |
| | **(e)** | Both the rates of the forward and backward reaction will be increased by the same extent when a catalyst is added. Therefore, the $K_p$ value does not change. | |
| | **(f)** | **(i)** | Positive and large. |
| | | **(ii)** | The ratio will be greater than 1. Since $\Delta G$ is negative, the reaction is spontaneous and the position of equilibrium lies to the right. |

| | | | |
|---|---|---|---|
| **3** | **(a)** | **(i)** | Nucleophilic substitution |
| | | **(ii)** | Warm **B** with aqueous sodium hydroxide, cool the mixture and add excess dilute nitric acid to acidify the solution. Then, add aqueous silver nitrate, followed by dilute ammonia solution. Formation of a white precipitate that dissolves in dilute ammonia solution to give a colourless solution confirms the presence of $Cl^-$ ions. This will further confirm that **B** is a chloroalkane. |
| | **(b)** | **(i)** | Ethanolic NaOH, heat under reflux. |
| | | **(ii)** | **D**<br><br>*trans* but-2-ene<br><br>**E**<br><br>*cis* but-2-ene<br><br>**F**<br><br>but-1-ene |
| | | **(iii)** | **D** and **E** are stereoisomers as both have different groups ($-CH_3$ and $-H$) bonded to each carbon atom of the C=C bond. **D** is a *trans* isomer as both H atoms are on opposite sides of the C=C bond, while **E** is a *cis* isomer as both H atoms are on the same side. **F** does not show stereoisomerism as carbon-1 of the C=C bond has 2 H atoms. |

| | | | |
|---|---|---|---|
| **4** | **(a)** | **(i)** | Step 1: conc $HNO_3$, conc $H_2SO_4$, maintained at 30 ºC<br>Step 2: 1. tin, conc HC*l*, heat; 2. NaOH(aq) |
| | | **(ii)** |  |
| | | **(iii)** | Constitutional isomerism. They are compounds with the same molecular formula but different structural formula. |
| | | **(iv)** | $C_9H_6N_2O_2$ |
| | **(b)** | **(i)** | Methyl group is 2,4-directing. The electron donating methyl group stabilises the arenium ions with the electrophile attacking at positions 2 and 4 rather than at position 3. Therefore, **H** is the least likely to be formed. |
| | | **(ii)** | The methyl group (which has a $sp^3$ hybridised carbon atom) sterically hindered the approach/attack of the electrophile at position 2 of the benzene ring.<br>(Stating that “repulsion occurred between the groups” lacks clarity.) |
| | **(c)** | **(i)** |  |

| | | | |
|---|---|---|---|
| | | **(ii)** | Benzene has a stable aromatic π electron cloud which will not favour addition reaction as it would result in the loss of aromaticity/resonance stability. Substitution will allow the benzene ring to retain aromaticity and resonance stability. |
| | | **(iii)** | <br>Methyl group is electron-donating and will increase electron density in the π electron cloud, activating the benzene ring towards electrophilic substitution. Therefore methylbenzene is more reactive than benzene. K has a $-COCH_3$ group which is electron-withdrawing and will reduce the electron density in the π electron cloud. This will make the aromatic ring less susceptible towards electrophilic attack. |

| | | | |
|---|---|---|---|
| **5** | **(a)** | **(i)** | Cell is operating at non-standard conditions where [$Cl^-$(aq)] >> 1 mol $dm^{-3}$, causing $E$(AgC*l*/Ag) to be more negative, and $O_2$ is dissolved instead of $O_2$(g) at 1 bar, causing $E(O_2/H_2O)$ becomes less positive. Since the magnitude of impact is larger on $E(O_2/H_2O)$ than E(AgC*l*/Ag), $E_{cell}$ becomes less positive than $E^{\ominus}_{cell}$. So when the external voltage is +1.45 V, there will be current flow.<br><br>(A small number of excellent responses appreciated that the cell operated under non-standard conditions, which resulted in the cell's electrode potential being lower than the standard cell potential.) |
| | | **(ii)** | Reduction. The oxidation number of oxygen decreased from 0 in $O_2$ to –2 in $H_2O$. |
| | | **(iii)** | [Show flow of electrons from silver electrode (ring-shaped anode) to platinum electrode (disk-shaped)] |
| | | **(iv)** | charge transferred in 1 min = 0.85 × 60 = 51 C<br><br>amount of electrons transferred in 1 min = 51 ÷ 96500 = $5.285 \times 10^{-4}$ mol<br><br>amount of $O_2$ reacted in 1 min = $5.285 \times 10^{-4} \div 4 = 1.321 \times 10^{-4}$ mol<br><br>rate of reaction = $1.321 \times 10^{-4} \times 6.02 \times 10^{23}$<br><br>= $7.95 \times 10^{19}$ molecules per minute |
| | | **(v)** | Negative. The number of aqueous ions decreased as the equation 1 proceeds, decreasing the number of ways of arranging the particles and their energy, resulting in lowered entropy/disorder. |
| | | **(vi)** | To ensure the oxygen gas in the water sample is well-mixed and concentration of oxygen remains constant at the electrodes during the measurement. |
| | | **(vii)** | Silver electrode (anode). The white solid to be removed is AgC*l*. |
| | | **(viii)** | • To keep the concentration of $Cl^-$ constant using a saturated solution so that the cell potential will only change due to a change in the concentration of $O_2$.<br>• Using a much lower concentration of $Cl^-$, equilibrium position of the reaction, $AgCl + e^- \rightleftharpoons Cl^- + Ag$, will shift to the right, resulting in a less negative $E$(AgC*l*/Ag) value. This will cause $E_{cell}$ to be less positive. |
| | **(b)** | **(i)** | Variable oxidation states (+2 and +3) |
| | | **(ii)** | $2s^22p^63s^23p^63d^4$ |

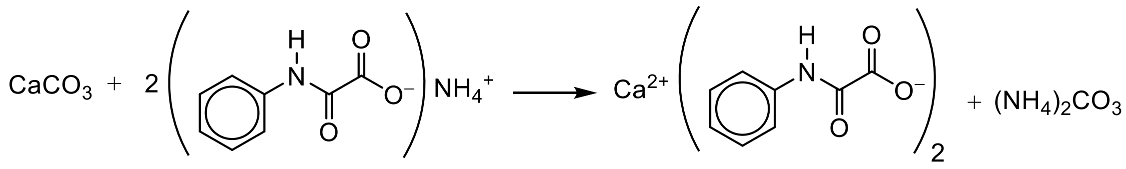
| | | | |
|---|---|---|---|
| | | **(iii)** | $I_2 \equiv 2\ Na_2S_2O_3 \equiv 2Mn(OH)_3 \equiv \frac{1}{2}O_2$<br>amount of $Na_2S_2O_3$ used = $0.100 \times \frac{11.20}{1000}$ = $1.12 \times 10^{-4}$ mol<br>amount of $I_2$ reacted = $1.12 \times 10^{-4} \div 2 = 5.60 \times 10^{-5}$ mol<br>amount of $O_2$ present = $5.60 \times 10^{-5} \div 2 = 2.80 \times 10^{-5}$ mol<br>concentration of $O_2$ = $2.80 \times 10^{-5} \div \frac{100}{1000}$ = $2.80 \times 10^{-4}$ mol dm$^{-3}$ |
| | **(c)** | **(i)** | amount of $O_2$ = $(8.24 - 6.93) \times 10^{-3} \div 32.0 = 4.09 \times 10^{-5}$ mol |
| | | **(ii)** | $pV = nRT$<br>$p(1 \times 10^{-3}) = (4.09 \times 10^{-5})(8.31)(273 + 35)$<br>$p = 1.048 \times 10^{2}$ Pa = 0.105 kPa<br>total pressure = 103.4 + 0.1 = 103.5 kPa |
| | | **(iii)** | $O_2(aq) \rightleftharpoons O_2(g)$<br>The oxygen gas already present would have shifted the position of equilibrium for water saturated with oxygen gas. This would have decreased the amount of oxygen gas that would escape out of distilled water due to Le Chatelier's Principle. The calculation in **(ii)** did not factor in how partial pressure of the oxygen gas will affect the position of equilibrium. |

**Paper 3**

**1 (a)** The thermal stability of the carbonates of Group 2 elements increases down the Group.

Down the Group, the radius of the metal cation, $M^{2+}$, increases and its charge density decreases. As a result, the ability of $M^{2+}$ to polarise the electron cloud of the large $CO_3^{2-}$ anion decreases and the C—O bonds are weakened to a smaller extent. Hence thermal stability of the Group 2 carbonates increases down the Group.

**(b) (i)**



**(ii)** Let x be the solubility of calcium *N*-phenyloxamate ($CaA_2$).

| | $CaA_2$(s) | $\rightleftharpoons$ | $Ca^{2+}$(aq) | + | $2A^-$(aq) |
|---|---|---|---|---|---|
| Equilibrium concentration / mol dm$^{-3}$ | | | x | | 2x |

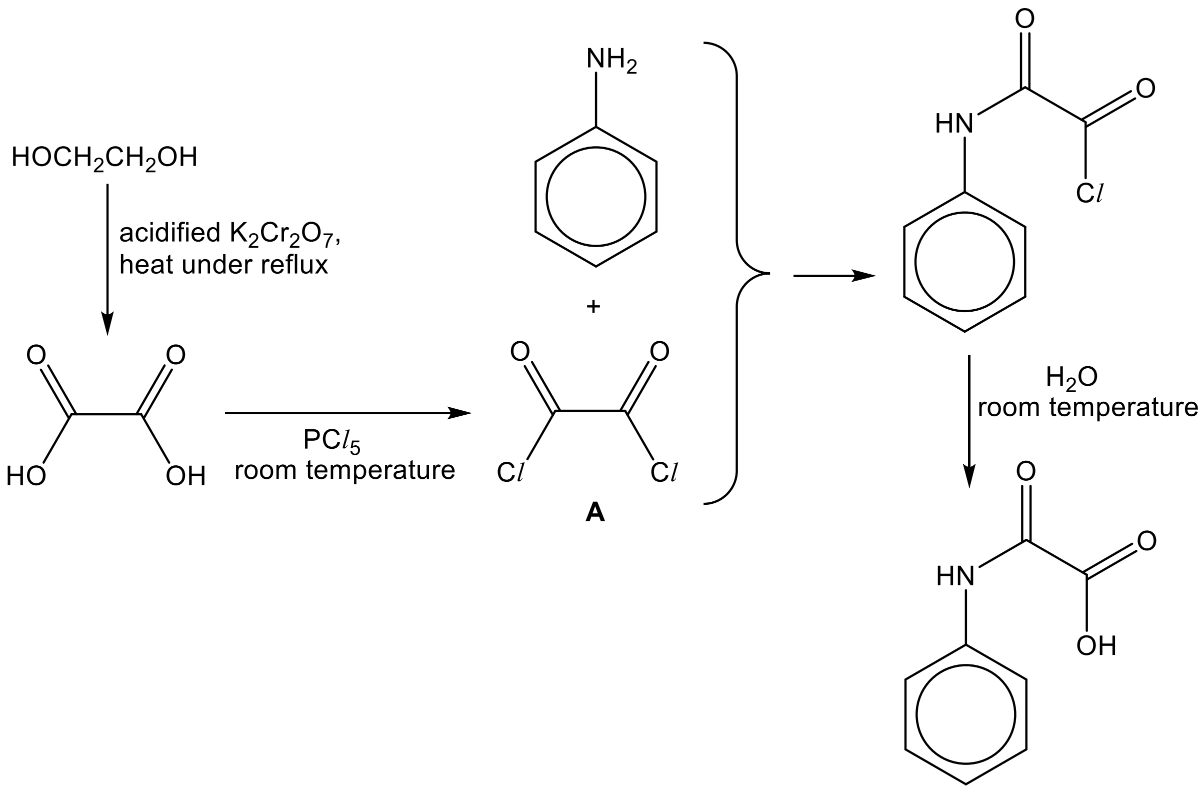
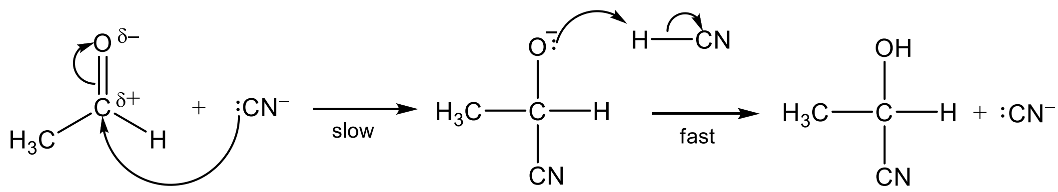
$(x)(2x)^2 = 1.75 \times 10^{-5}$

$x = 1.636 \times 10^{-2}$ mol dm$^{-3}$

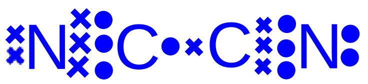
Concentration of calcium N-phenyloxamate in a saturated solution

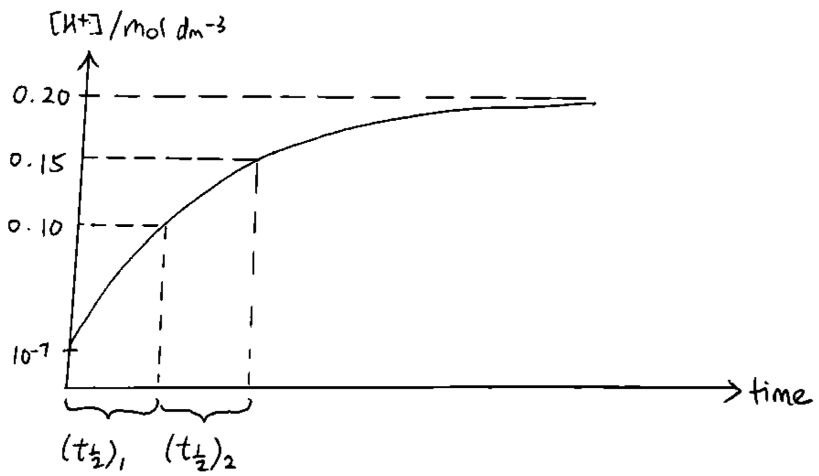
$= 1.636 \times 10^{-2} \times [40.1 + 2 \times (8 \times 12.0 + 6 \times 1.0 + 14.0 + 3 \times 16.0)]$

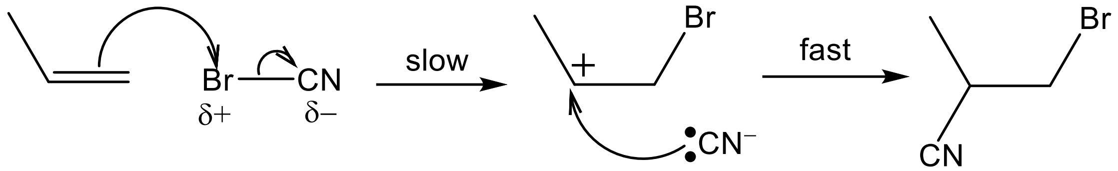
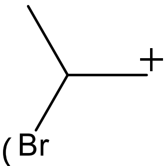
$= 6.02$ g dm$^{-3}$

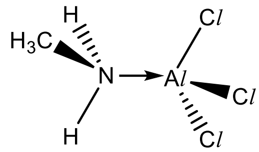
| | | | |
|---|---|---|---|
| **(c)** | |  | |
| **(d)** | | Amides are neutral because the p orbital of N atom overlaps with the $\pi$ electron cloud of the neighbouring carbonyl group (C=O). The lone pair of electrons on N atom is delocalised into the C=O group and hence is not available for protonation. | |
| **(e)** | **(i)** | $HCN + OH^- \rightarrow :CN^- + H_2O$ <br>  | |
| | **(ii)** | 1,3-butadiene does not undergo nucleophilic addition. <br><br> This is because nucleophiles will not be attracted to the C=C bond (OR will be repelled by the $\pi$ electrons in the C=C bond) as it is electron rich and the donation of a lone pair of electrons from the nucleophiles will not be favoured. <br><br> For 4-methyl-1-penten-3-one, on the C=O group, the O is electronegative and the C=O bond is polarised, the $:CN^-$ nucleophile will be attracted to the electron deficient carbon of the carbonyl group. <br><br> Due to the steric hindrance of the alkene group and the isopropyl group [—$CH(CH_3)_2$], the nucleophilic attack on C=O will be diminished and **B** will not be formed. | |

| | | | |
|---|---|---|---|
| | | | As the C═O carbon and the alkene carbons are all sp$^2$ hybridised, the $\pi$ electrons are extensively delocalised across the three sp$^2$ carbon and oxygen atoms. Likewise, the $\delta$+ charge on carbon-3 can also delocalise over to carbon-1. This will allow the less sterically hindered carbon-1 to undergo nucleophilic attack by :$CN^-$ to form **C**.<br><br>(Note: The actual mechanism is different from the suggested answer above. The actual mechanism involves the formation of a carbocation (instead of a carbon with partial positive charge) and tautomerisation, which is not in the H2 Chemistry syllabus.) |

| | | | |
|---|---|---|---|
| **2** | **(a)** | **(i)** | <u>Reactants</u><br>C: –4<br>N: –3<br>O: 0<br><br><u>Products</u><br>C: +2<br>N: –3<br>O: –2 |
| | | **(ii)** | $\Delta H$ = [8(+410) + 6(+390) + 3(+496)] – [2(+410) + 2(+890) + 12(+460)]<br>= –1012 kJ mol$^{-1}$ |
| | | **(iii)** | $K_a = \frac{[H^+][CN^-]}{[HCN]}$<br>$\frac{(10^{-10})[CN^-]}{[HCN]} = 7.2\times10^{-10}$<br>$\frac{[CN^-]}{[HCN]} = 7.20$ |
| | **(b)** | **(i)** | Any reagent that has a reduction potential that is more positive than +0.37 V:<br>Acidified $KMnO_4$<br>$2MnO_4^- + 6H^+ + 10HCN \rightarrow 5C_2N_2 + 8H_2O + 2Mn^{2+}$<br>$E^\ominus_{cell}$ = +1.52 – (+0.37) = +1.15 V |
| | | **(ii)** | <br>sp hybridisation |
| | | **(iii)** | The C≡N bonds in cyanogen are polar as N is more electronegative than C. The C–C bond is non-polar as both C atoms have the same electronegativity. Since the molecule is linear, both dipole moments due to the two C≡N bonds cancel out, leading to net dipole moment of zero. Therefore, cyanogen is a non-polar molecule. |
| | **(c)** | **(i)** | Hydrolysis or Nucleophilic substitution |

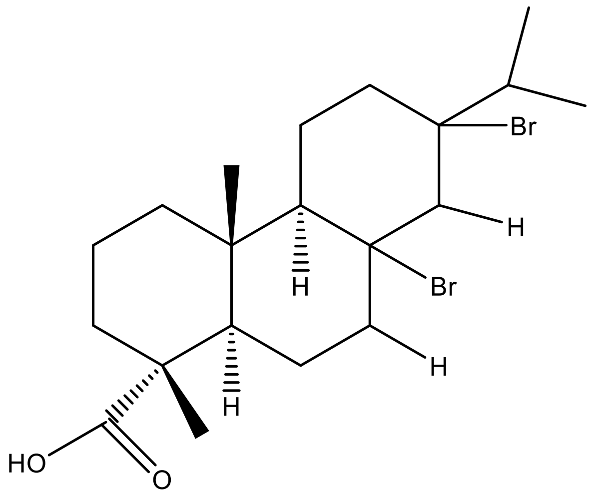
| | | | |
|---|---|---|---|
| | | **(ii)** | Water is present in large excess. The concentration of water will remain relatively constant as compared to the concentration of the other reactant. Therefore, concentration of water will not appear in the rate equation and the reaction will be pseudo first-order with respect to C*l*CN. |
| | | **(iii)** | Nucleophile. The lone pair of electrons on O of water is donated to the electron deficient carbon in cyanogen chloride.<br>(Few candidates were able to provide an explanation.) |
| | | **(iv)** | rate = $k$[C*l*CN][$OH^-$] |
| | | **(v)** | [C*l*CN] = $0.010 \div \frac{100}{1000}$ = 0.100 mol $dm^{-3}$<br>[$OH^-$] = $10^{-4}$ mol $dm^{-3}$<br>$\frac{rate_2}{rate_3} = \frac{5.1\times10^{-7}\times0.100}{4.2\times0.100\times10^{-4}} = 1.20 \times 10^{-3}$ |
| | **(d)** | | <u>Continuous method of measuring rate of reaction 2</u><br>Start the stopwatch when 0.010 mol of C*l*CN is added to 100 $cm^3$ of water. Assuming that HOCN is a strong acid, the total concentration of $H^+$ will be 0.20 mol $dm^{-3}$.<br>Measure the pH of the solution using a pH meter throughout the experiment. [$H^+$] can be calculated from pH.<br>Plot a graph of [$H^+$] against time. From this [product]-time graph, the half-life can be determined by measuring the time taken required for concentration of [$H^+$] to go from $10^{-7}$ mol $dm^{-3}$ *i.e.* $[H^+]_0$ to 0.10 mol $dm^{-3}$ *i.e.* ½$[H^+]_{max}$ and from 0.10 mol $dm^{-3}$ *i.e.* ½$[H^+]_{max}$ to 0.15 mol $dm^{-3}$ *i.e.* ¾$[H^+]_{max}$.<br>Using the expression, $k = \frac{\ln 2}{t_{\frac{1}{2}}}$ , $k$ can be calculated.<br><br> |

| | | |
|---|---|---|
| | **(e)** | Electrophilic addition<br><br>In the electrophilic addition mechanism, the δ+ Br atom of the polar BrCN molecule attacks the C=C bond to form the carbocation. Since the secondary carbocation is more stable than the primary carbocation (  ), isomer **D** will be formed rather than isomer **E**. |

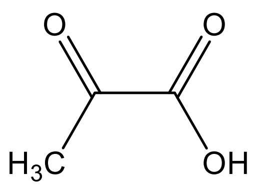
| | | | |
|---|---|---|---|
| **3** | **(a)** | **(i)** | Amines can act as a Lewis base due to the lone pair of electrons on its N atom that is available for donation. According to the Lewis theory of acids and bases, a base is an electron pair donor.<br><br>$CH_3NH_2 + AlCl_3 \rightarrow$  (3D structure not required)<br><br>The lone pair of electrons on the N atom in methylamine is donated to the electron deficient A*l* in $AlCl_3$. |
| | | **(ii)** | Order of basicity<br>trimethylamine > dimethylamine > methylamine<br><br>The larger the number of electron-donating alkyl groups, the more available the lone pair of electrons on N for donation. Therefore, trimethylamine is the stronger Lewis base, followed by dimethylamine, and methylamine is the weakest Lewis base. |
| | **(b)** | **(i)** | $K_b = \frac{[CH_3CH_2NH_3^+][OH^-]}{[CH_3CH_2NH_2]}$ |
| | | **(ii)** | Ethylamine is the strongest base due to the electron-donating ethyl group that will make the lone pair of electrons on N atom more available for protonation/ donation than ammonia, which does not have an alkyl group.<br><br>Phenylamine is less basic than ammonia as the orbital containing the lone pair of electrons on N overlaps with the $\pi$ electron cloud of benzene. Since the lone pair of electrons is delocalised into the benzene ring, it is less available for protonation/ donation.<br><br>4-chlorophenylamine is the least basic as the electron-withdrawing chlorine atom increases the delocalisation of the lone pair of electrons on N atom of $-NH_2$ group into the benzene ring, hence making the lone pair of electrons on N atom less available for protonation/ donation. |
| | **(c)** | **(i)** | $[Cu(H_2O)_6]^{2+} + 4NH_3 \rightarrow [Cu(NH_3)_4(H_2O)_2]^{2+} + 4H_2O$<br>Ligand exchange |

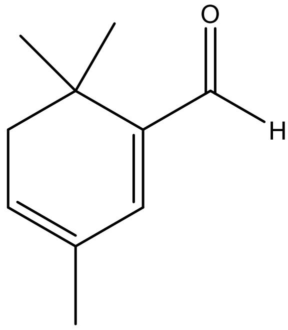
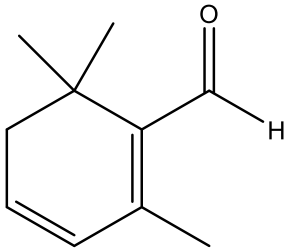
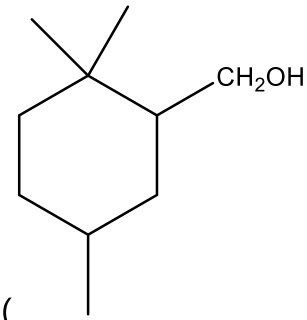
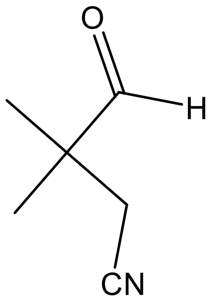
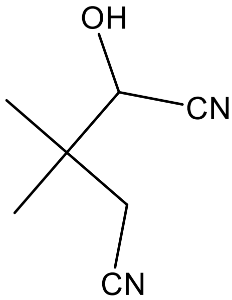
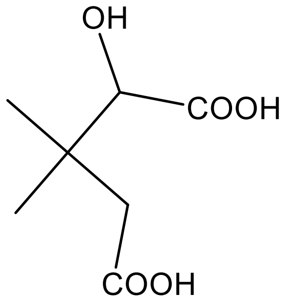
| | | | |
|---|---|---|---|
| | | **(ii)** | $[Cu(NH_3)_4(H_2O)_2]^{2+}$ + 2***en*** $\rightarrow$ $[Cu(\textbf{\textit{en}})_2(H_2O)_2]^{2+}$ + $4NH_3$<br><br>$CuSO_4$(aq) contains $Cu^{2+}$ which has a partially filled 3d subshell (electronic configuration $1s^22s^22p^63s^23p^63d^9$) that shows d-d transition in the presence of $NH_3$ and $H_2O$ ligands. (The 3d orbitals are split into two sets of orbitals with different energies. The difference in energies ($\Delta E$) between these two sets of non-degenerate 3d orbitals is small and radiation from the visible region of the electromagnetic spectrum is absorbed when an electron is promoted from a lower energy d orbital to another unfilled/partially-filled d orbital of higher energy.) In the presence of bidentate ***en*** ligands *i.e.* $[Cu(H_2O)_6]^{2+}$, ligand exchange takes place and the $\Delta E$ changes to a larger value*. Therefore, the complement of the absorbed colour changes due to the different $\Delta E$. The colour observed also changes from blue to purple.<br><br>*Note to students:<br><br>From: $\Delta E = h\nu = hf = \frac{hc}{\lambda}$<br><br>$\Delta E$ is inversely proportional to $\lambda$, the wavelength of the radiation absorbed.<br><br>When the colour observed changes from blue to purple (or violet), the main complementary colour absorbed changes from orange to yellow.<br><br>This means that $\lambda$ from orange to yellow decreases (~650 nm to ~580 nm).<br><br>Hence $\Delta E$ increases, *i.e.* changes to a larger value. |
| | **(d)** | All four compounds have similar $M_r$ and would have similar instantaneous dipole-induced dipole (id-id) interactions between molecules. Glycine has the highest melting point as it exists as a zwitterion and has giant ionic lattice structure. Most energy is required to overcome the strong electrostatic attraction between oppositely charged ions.<br><br>Both butylamine and propanoic acid are polar molecules with simple covalent structures. There is more extensive hydrogen bonding between propanoic acid molecules than butylamine molecules as it has more sites for formation of hydrogen bonds. In addition, the O–H bond in propanoic acid is more polar than the N–H bond in butylamine due to the presence of the electron-withdrawing carbonyl group and the more electronegative O atom. Hence, more energy is required to overcome the more extensive and stronger hydrogen bonds between propanoic acid molecules than that between butylamine molecules.<br><br>Pentane is a non-polar molecule with simple covalent structure. Least energy is required to overcome the weak id-id forces of attraction exist between molecules. Hence, pentane will have the lowest melting point. | |

| | | | |
|---|---|---|---|
| **4** | **(a)** | **(i)** | The reactivity of halogens as oxidising agents decreases down the Group. As observed from the $E^\ominus$ values below,<br><br>**Element** / **$E^\ominus$ / V**:<br>$\frac{1}{2}F_2 + e^- \rightleftharpoons F^-$ : +2.87<br>$\frac{1}{2}Cl_2 + e^- \rightleftharpoons Cl^-$ : +1.36<br>$\frac{1}{2}Br_2 + e^- \rightleftharpoons Br^-$ : +1.07<br>$\frac{1}{2}I_2 + e^- \rightleftharpoons I^-$ : +0.54<br><br>The more positive the $E^\ominus$ value, the easier for the halogen molecule to be reduced and the higher its reactivity as an oxidising agent. |
| | | **(ii)** | Halogen displacement reaction: $Cl_2$ is able to displace $Br^-$ to form $Br_2$ ($E^\ominus_{cell}$ = +0.29 V) but $Br_2$ is unable to displace $Cl^-$ to form $Cl_2$ ($E^\ominus_{cell}$ = −0.29 V). This shows that $Cl_2$ is a stronger oxidising agent than $Br_2$. |
| | **(b)** | **(i)** | $MgCl_2$(s) dissolves readily in water to form a weakly acidic solution of pH = 6.5.<br>$MgCl_2(s) + 6H_2O(l) \rightarrow [Mg(H_2O)_6]^{2+}(aq) + 2Cl^-(aq)$<br>$[Mg(H_2O)_6]^{2+}(aq) + H_2O(l) \rightleftharpoons [Mg(H_2O)_5(OH)]^+(aq) + H_3O^+(aq)$<br><br>$AlCl_3$(s) dissolves readily in water to form an acidic solution of pH = 3.<br>$AlCl_3(s) + 6H_2O(l) \rightarrow [Al(H_2O)_6]^{3+}(aq) + 3Cl^-(aq)$<br>$[Al(H_2O)_6]^{3+}(aq) + H_2O(l) \rightleftharpoons [Al(H_2O)_5(OH)]^{2+}(aq) + H_3O^+(aq)$<br><br>$SiCl_4$(l) dissolves in water to form a strongly acidic solution of pH = 2 and a white crystalline solid $SiO_2$.<br>$SiCl_4(l) + 2H_2O(l) \rightarrow SiO_2(s) + 4HCl(aq)$ |
| | | **(ii)** | $MgCl_2$ has a giant ionic lattice structure and the compound undergoes dissolution and hydration of ions. Partial hydrolysis of $Mg^{2+}$(aq) occurs to give a slightly acidic solution.<br><br>$SiCl_4$ has a simple covalent structure and undergoes hydrolysis in water to form $HCl$ that gives an acidic solution. This is because Si atom in $SiCl_4$ has energetically accessible vacant 3d orbital for dative bonding with water molecules. |
| | **(c)** | **(i)** | $K_c = \dfrac{[H^+][X^-][HXO]}{[X_2]}$<br><br>Units: $mol^2\ dm^{-6}$ |
| | | **(ii)** | $I_2 \quad + \quad 2OH^- \quad \rightleftharpoons \quad IO^- \quad + \quad I^- \quad + \quad H_2O$ |

| | | | |
|---|---|---|---|
| | | | I 0.10 0.5 0 0<br>C –0.097 –0.194 +0.097 +0.097<br>E 0.003 0.306 0.097 0.097<br>$K_c = \dfrac{0.097\times0.097}{0.003\times(0.306)^2}$ = 33.5 mol$^{-1}$ dm$^3$ |
| | **(d)** | **(i)** | <br>2 extra chiral centres |
| | | **(ii)** | KC*l* has a giant ionic lattice structure with strong electrostatic forces of attraction between the oppositely charged $K^+$ and $Cl^-$ ions that require a lot of energy to overcome. As a result, it will have a high boiling point and will not be volatile. |
| | | **(iii)** | formula of tetranitratoethane is $C_2H_2O_{12}N_4$<br>mass percentage of oxygen = $\dfrac{12\times16.0}{2\times12.0+2\times1.0+12\times16.0+4\times14.0}$ = 70.1% |
| | | **(iv)** | $7C(s) + 2C_2H_2O_{12}N_4(s) \rightarrow 11CO_2(g) + 2H_2O(g) + 4N_2(g)$ |
| | | **(v)** | Any two of the following:<br>• Formation of a large number of moles of products which increase the number of ways of arranging the particles, resulting in greater entropy/ disorder.<br>• Formation of gaseous products from solid reactants/ no non-volatile products (which has high degree of order) is formed. Gases have greater entropy as there are greater ways to arrange the particles as compared to solids where the particles are regularly arranged in a lattice structure. |

| | | | |
|---|---|---|---|
| | | | • As the reaction is exothermic, there is an increase in entropy due to the larger number of ways to distribute/ arrange the different quanta of energy in the system. |

| | | | |
|---|---|---|---|
| **5** | **(a)** | **(i)** | Transition metal complexes contains transition metal ions which has a partially filled d subshell. In the presence of ligands, the d orbitals are split into two sets of orbitals with different energies. The difference in energies ($\Delta E$) between these two sets of non-degenerate d orbitals is small and radiation from the visible region of the electromagnetic spectrum is absorbed when an electron is promoted from a lower energy d-orbital to another unfilled/partially-filled *d* orbital of higher energy. The colour observed corresponds to the complement of the absorbed colours. Hence, the complexes are coloured. |
| | | **(ii)** | The vanadium ion ($V^{3+}$) in $V_2(SO_4)_3$ has an electronic configuration of $1s^22s^22p^63s^23p^63d^2$ whereas the vanadate ion in $NaVO_3$ has vanadium with an electronic configuration of $1s^22s^22p^63s^23p^63d^0$. Only the vanadium in $V_2(SO_4)_3$ has a partially filled 3d subshell that can have d-d transition and will be coloured. The vanadium in $NaVO_3$ has no electrons in the 3d subshell and will be colourless. |
| | **(b)** | **(i)** | $VO_3^- + 4H^+ + e^- \rightleftharpoons VO^{2+} + 2H_2O$ --- reaction 1<br>$VO_2^+ + 2H^+ + e^- \rightleftharpoons VO^{2+} + H_2O$ --- reaction 2<br>As pH increases, $[H^+]$ decreases. Therefore, as $[H^+]$ is lowered, the positions of equilibrium of both reaction 1 and equation 2 shift to the left, making both of them weaker oxidants as pH increases.<br><br>At pH = 7, as there is a larger number of moles of $H^+$/more $H^+$ terms in equation for reaction 1, position of equilibrium will lie on the left to a greater extent, resulting in a more negative $E(VO_3^-/VO^{2+})$ value. Therefore, $VO_2^+$ is a more powerful oxidant. |
| | | **(ii)** | $E^\ominus(VO_3^-/VO^{2+}) = +1.00$ V<br>$E^\ominus(SO_4^{2-}/SO_2) = +0.17$ V<br>$E^\ominus_{cell} = (+1.00) - (+0.17)$<br>$= +0.83$ V $> 0$<br>$VO_3^-$ is able to oxidise $SO_2$ to $SO_4^{2-}$, itself forming $VO^{2+}$.<br>$2VO_3^- + 4H^+ + SO_2 \rightarrow SO_4^{2-} + 2VO^{2+} + 2H_2O$<br>(Candidates were able to predict the products of the reactions by considering the electrode potential values.) |
| | **(c)** | **(i)** |  |

| | | | |
|---|---|---|---|
| | | **(ii)** |   |
| | | **(iii)** | $C_{10}H_{20}O$ (  ) |
| | **(d)** | **(i)** | • Since **J** does not react with 2,4-DNPH, **J** does not have a carbonyl functional group<br>• Since **J** does not react with bromine water, **J** does not have an alkene functional group<br>• Since **J** reacts with sodium metal and is neutral, **J** cannot have a carboxylic functional group. It should have an alcohol group. |
| | | **(ii)** | **H**  **J**  **K**  |
| | | **(iii)** | Step 1:<br>Ethanolic KCN, heat under reflux<br><br>Step 3:<br>dilute HC*l*(aq)/ $H_2SO_4$(aq), heat under reflux<br>Step 4:<br>Acidified $K_2Cr_2O_7$, heat (under reflux) |